

Khulna University of Engineering & Technology
 Department of Building Engineering and Construction Management
 B. Sc. Engineering 2nd Year 1st Term Regular Examination, 2017
 Math 2123
 (Mathematics-III)

Full Marks: 210

Time: 3 hrs

- N.B. i) Answer any three questions from each section in separate script.
 ii) Figures in the right margin indicate full marks.

Section - A

- (a) Define integral transform and kernel of transform. Also find the kernel of Laplace transform. (10)
- (b) Find the Laplace transform of $te^{2t} \sin 3t$. (10)
- (c) Solve the following I.V. P. by Laplace transform (15)
- $Y'' + 9Y = \cos 2t, Y(0) = 1, Y(\pi/2) = -1$
- Find the inverse Laplace transform of $\frac{11s^2 - 2s + 5}{(s-2)(2s-1)(s+1)}$ Using Heaviside expansion formula. (10)
- (b) A cantilever beam is clamped at the end $x = 0$ and is free at the end $x = L$. It carries a load per unit length given by
- $w(x) = \begin{cases} w_0, & 0 < x < L/2 \\ 0, & L/2 < x < L \end{cases}$
- Find the deflection. Using Laplace transform
- (c) In each determine whether the vectors are linearly independent or linearly dependent (10)
- $A = 2\hat{i} + \hat{j} - 3\hat{k}, B = \hat{i} - 4\hat{k}, C = 4\hat{i} + 3\hat{j} - \hat{k}$
 $A = \hat{i} - 3\hat{j} + 2\hat{k}, B = 2\hat{i} - 4\hat{j} - \hat{k}, C = 3\hat{i} + 2\hat{j} - \hat{k}$
3. (a) If $A = \frac{r}{r}$ find grad div A (08)
- (b) Find an equation for the tangent plane to the surface $2xz^2 - 3xy - 4z = 7$ at the point $(1, -1, 2)$ (12)
- (c) A particle moves along a curve whose parametric equations are $x = e^{-t}, y = 2 \cos 3t, z = 2 \sin 3t$, where t is the time. Find the magnitudes of the velocity and acceleration at time, $t = 0$. (15)
4. (a) Test the vector $A = (6xy + z^3)\hat{i} + (3x^2 - z)\hat{j} + (3xz^2 - 4)\hat{k}$, is irrotational or not. If the vector is irrotational find its scalar potential. (13)
- (b) Find the Directional derivative of $P = 4e^{2x-y+z}$ at the point $(1, 1, -1)$ in a direction toward the point $(-3, 5, 6)$. (10)
- (c) If $A = (2y + 3)\hat{i} + xz\hat{j} + (yz - x)\hat{k}$, evaluate $\int_C A \cdot d\vec{r}$ along the following path C: the straight lines from $(0, 0, 0)$ to $(0, 0, 1)$ then to $(0, 1, 1)$, and then to $(2, 1, 1)$. (12)

Section - B

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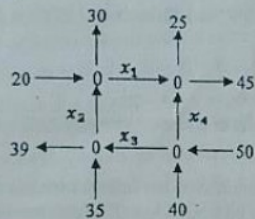
- 5 (a) Define with example (i) skew-symmetric matrix, (ii) principal sub matrix, elementary matrix, (iv) diagonal matrix.
- (b) Express the matrix A in the form $P+Q$, where P is a symmetric matrix and Q is a skew-symmetric matrix;
- $$A = \begin{bmatrix} 3 & 3 & -1 \\ 0 & 3 & -2 \\ -1 & 2 & 2 \end{bmatrix}$$
- (c) Reduce the matrix to its echelon form then to its canonical form then to its normal form. And also find its rank where (15)

$$A = \begin{bmatrix} 2 & 1 & 5 & 1 & 5 \\ 1 & 1 & -3 & -4 & -1 \\ 3 & 6 & -2 & 1 & 8 \\ 2 & 2 & 2 & -3 & 2 \end{bmatrix}$$

- 6 (a) Find the inverse of the matrix $A = \begin{bmatrix} 0 & 2 & 1 & 3 \\ 1 & 1 & -1 & -3 \\ 1 & 2 & 0 & 1 \\ -1 & 1 & 2 & 6 \end{bmatrix}$ using elementary transformation. (15)

- (b) What do you mean by a consistent system of linear equations? What are the different method of solving non-homogenous system of linear equations? Investigate for consistency of the following equations and if possible find the solutions of $4x - 2y + 6z = 8$, $x + y - 3z = -1$, $15x - 3y + 9z = 21$ by using matrix. (20)

- 7 (a) Find the traffic flow in the adjoining network of one way streets. Is the solution unique? (13)



- (b) Define eigen value and eigen vector of a matrix. Find all eigen values and associated eigen vectors of the following matrix: $A = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 0 & 0 \\ -2 & 0 & 4 \end{bmatrix}$ (22)

- 8 (a) Find $\text{adj } A$ and hence A^{-1} , where $A = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ (10)

- (b) Verify the Cayley-Hamilton theorem for the matrix $\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ (10)

- (c) Find the nonsingular matrices P & Q such that PAQ is normal form where, (10)

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 1 & 3 & 2 \\ 2 & 1 & 3 \end{bmatrix}$$